Isha Kulshrestha

B.Sc. (Hons) Physics Undergraduate (4-year program)
Lovely Professional University, Punjab, India
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Research Interests

Physical acoustics, wave propagation, aeroacoustics, signal processing, computational modeling of acoustic systems, environmental noise characterization.

Education

B.Sc. (Hons) Physics 2023–2027
Lovely Professional University, Punjab CGPA: **8.51/10**
Relevant Coursework: Waves and Optics, Mathematical Physics, Classical Mechanics, Electromagnetic Theory, Differential Equations, Multivariable Calculus, Statistical Mechanics, Classical Dynamics.

Higher Secondary (Class XII) 2023
Delhi Public School, Dehradun 86.66%

Secondary (Class X) 2021
Delhi Public School, Dehradun 92.8%

Research Experience

Summer Research Intern Jun 2026 – present
Department of Mechanical Engineering, NIT Calicut

- Developed a Python-based signal processing pipeline to characterise propeller noise across an RPM sweep of 2250–5700, identifying blade passage frequency and SPL from averaged FFT spectra.
- Trained a linear regression model on spectral measurements to estimate rotor RPM, with extrapolation evaluated up to 10,000 RPM.

Research Projects

Comparative Analysis of Acoustic Barrier Insertion Loss 2025–present
github.com/IshaK-2006/Acoustic-Barrier-Diffraction

- Developed 2D FDTD simulation (Python) for sound propagation over rigid ground; conducted field experiments over grassy ground.
- **Key finding:** Experimental interference pattern showed systematic peak/dip inversion relative to rigid-ground simulation — consistent with $\sim 180°$ phase shift from porous ground surface.

Lombard Effect: Acoustic Signal Analysis of Noise-Induced Vocal Modulation
github.com/IshaK-2006/lombard-effect

- Controlled experiment studying adaptive vocal modulation under varying ambient noise conditions.
- Performed spectral analysis (baseline-corrected SPL, fundamental frequency extraction) using Audacity and Python.
- Applied regression modeling to quantify the relationship between environmental noise level and vocal response.

Computational Modeling of Coupled Oscillatory Systems
github.com/IshaK-2006/small_oscillations

- Formulated equations of motion for multi-degree-of-freedom systems in matrix form.
- Reduced dynamics to a generalized eigenvalue problem; implemented numerical solvers in C++.
- Analyzed eigenfrequency spectra and mode structures under systematic parameter variation.

Research Training

Bioacoustics Winter School Jan 2026
Université Jean Monnet Saint-Étienne (ENES lab), France *Mention Bien* (with distinction)

- Selective in-person international program; only Bachelor's-level participant among an otherwise postgraduate and postdoctoral cohort.
- Intensive training in time-frequency analysis, ecoacoustics, animal acoustic communication, and statistical modeling of biological signals.
- Completed assessed practical assignments in spectral feature extraction and quantitative acoustic signal analysis.
- Tools: Python, R, Praat, Audacity.

Technical Skills

Programming: Python (NumPy, SciPy, Matplotlib, Librosa), C++, R, LaTeX
Signal Analysis: Spectral analysis, FFT, time-frequency methods, SPL measurement, regression modeling
Tools: Audacity, Praat, Jupyter Notebook
Mathematical Methods: Linear algebra, eigenvalue problems, ODEs/PDEs, numerical modeling

Certifications

- **Fundamentals of Audio and Music Engineering: Part 1 — Musical Sound & Electronics**
  University of Rochester (Coursera), 2025
- **Intro to Acoustics (Part 1)**
  KAIST (Coursera), 2024

Additional

- **IELTS Academic:** Band 8.0 Nov 2025
- **Hindustani Classical Music:** Trained since 2016; taught music to children from underserved communities in Dehradun.
- Event Organizer & Fund Manager, Astrophysics Club (Uttara), LPU
- Community Development Intern, Samadhan NGO, Dehradun (2024)